

Description

This dataset contains 336 geolocated Lightning Network topology snapshots covering the years 2019–2023. The snapshots were reconstructed from Decker’s publicly available gossip message data, accessible at <https://github.com/lnresearch/topology> (last snapshot accessed on 2023-09-24).

All files are bundled in a single archive: `snapshots.geo.zip`. Each snapshot is stored in *GML* format and named using the pattern `YYYYMMDD.gml.geo` (e.g., `20190120.gml.geo`), where the filename corresponds to the snapshot date. Snapshots can be loaded directly using the standard `nx.read_gml()` function from the *NetworkX* library.

The archive also includes `shapes.geo.csv`, which provides metadata for each snapshot, including:

- Unix timestamp and corresponding datetime
- Number of nodes and channels
- Average node degree
- Approximated graph diameter
- Number of geocoded nodes
- Associated snapshot filename

Example:

```
timestamp,datetime,nodes,channels,degree,diameter,geocoded_nodes,file_name
1547942400,2019-01-20,1137,3240,5.699208443271768,7,771,20190120.gml.geo
1548028800,2019-01-21,1222,3760,6.153846153846154,7,826,20190121.gml.geo
```

A working demonstration of how to load and analyze the data is provided in the `example.ipynb` notebook.

Note: The archive also contains `scripts.zip`, which includes the reconstruction scripts for archival purposes only. It is recommended to use the actively maintained version available at: <https://github.com/ellariel/ln-data-preparation>

If you use this dataset, please cite it appropriately, along with the original Decker topology datasets.

Sources and References

- Decker, C. (2020). *Lightning Network Research – Topology Datasets*. DOI: 10.5281/zenodo.4088530 URL: <https://github.com/lnresearch/topology>
- IPinfo Developer API (used for geolocation): <https://ipinfo.io>